Fingerprint Reasoner: Theoretical Analysis of Raman Spectroscopy and Knowledge-Augmented Reasoning for Organic Compound Identification with Large Language Models

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Abstract

1 Introduction

Raman spectroscopy [1] is a non-invasive analytical technique based on molecular vibrations that provides rich chemical structure information. Quantum theory indicates that the molecular structures (chemical bonds and their spatial relations) are represented in vibrational modes of the molecules, and thus can be represented through raman spectroscopy. The spectrum of each compound has a characteristic "fingerprint" peak or region that can be used for identification and quantitative analysis [2]. Compared with other analytical techniques such as Mass Spectrometry (MS), Nuclear Magnetic Resonance (NMR), Infrared Spectroscopy (IR), and Ultraviolet-Visible Spectroscopy (UV-Vis), Raman spectroscopy has significant advantages that it is fast, non-destructive, requires little sample preparation, works well in water, offers high chemical specificity, and enables microscopic and in situ analysis [3]. For decades Raman spectroscopy had a limited application because of the very low efficiency of raman scattering and the costly devices which are not suiteable for onsite analysis [4]. With the development of low-cost, portable

apparatus using surface-enhanced Raman spectroscopy (SERS) [5], Raman spectroscopy has shown broad prospects in various fields [6]. Previous works with spectroscopy include tracking the process of chemical reactions [7], cancer diagnostics [8], measuring concentrations of biomolecules [6] or gas in disease diagnostics [9], or assisting the design of metal monatomic catalysts [10]. However, it is extremely difficult to identify molecular structures accurately with spectral data [6]. Even worse, due to lack of large-scale labeled data and the fact that raman spectrum is dependent on the environment, the performance of existing methods is limited, and previous works mainly focus on categorizing or predicting structure in very specific area, and have insufficient genaralization ability.
In recent years, Machine learning methods have given promising prospects of achieving breakthroughs. Rescarchers have applied deep learning methods in predicting spectrum of certain molecules [11]. However, its inverse process, identification using spectroscopic measurements, holds greater practical value and scientific significance. Previous works [12] on this topic are only effective on identifying functional groups, and have low accuracy rates on molecule identification. Solving spectroscopic analysis problems is challenging due to the quantum nature of raman spectroscopy, and it demands a high level of abstraction to understand the finger-print incorporated in peak positions. Then we raise the research problem: *Can LLMs master the basics of Quantum Physics and Spectroscopy and identify organic compounds with high accuracy?*

To solve this problem, we starts by analyzing the common errors, which reveal two primary factors behind the frequent failures. Namely, (1) LLMs suffer from hallucination, especially when the input data is insufficient, so it often generates completely absurd results, ignoring all given information. Even sometimes it begins to iterate through non-organic ionic compounds, though it is given that the input is spectrum of organic molecule. (2) LLMs often lack knowledge of basic empirical rules in the correlation of special bonds and peak regions, and often make fallacious assumptions.

To mitigate these problems, we propose **Fingerprint Reasoner**, a novel approach leveraging Large language models (LLM), which can be used to analyze insufficient data and extract useful features just as a human expert would do. Problem (1) mainly derives from the fact that LLMs have too much choices of compounds, so there are hardly any entry points of thought. Problem (2), on the other hand, is caused by insufficient related data in the training set. For problem (1), we designed an algorithm to select candidates with similar spectral patterns, which is based on the fact that molecules with similar structures have similar spectra. For problem (2), we designed a workflow, which contains knowledge retrieval and checking, incorporating the Chain of Thought (CoT) method [13] and the idea of knowledge-augmented agent for reasoning problems [14].

Our key contributions are summarized as follows:

- We identify two key limitations of solving spectroscopy problems with LLMs.
- We propose a novel method called Fingerprint Reasoner, which combines LLMs with knowledge retrieval and CoT.
- We provide a theoretical analysis of Raman spectroscopy, and reproduced optical apparatus to obtain spectral data as well as extracted the features of different categories of data, which supports our design of the method.
- We conduct extensive experiments on QM9S dataset, demonstrating that our method achieves high accuracy.

This work provides a new perspective for the application of LLM in the field of spectroscopy, and also enlightens the prospect for on-site and real-time identification of organic compounds, which has great potential in chemstry, biology, physics, medicine, and a lot more fields.

2 Preliminaries

2.1 Quantum theory

Raman Scattering is the inelastic scattering of a photon. To be exact, a small fraction of the photon are scattered by an excitation, whose frequency slightly differ from the incident photons. The process can be analyzed with time-dependent perturbation theory [15]. Here we consider the scattering of photons by the electron cloud of a molecule. Before scattering, the atom is in state $|i\rangle$, and after scattering, the atom is in state $|f\rangle$. The energy of the incident photon and scattering photon is ϵ and ϵ'.
The process can be described as subsystem A ($|a\rangle$) transitions to $|b\rangle$, during which in subsystem B the first mode loses one excitation and the second mode gains one:

$$|i\rangle = |a\rangle \otimes |n_1, n_2\rangle \, ; |f\rangle = |b\rangle \otimes |n_1 - 1, n_2 + 1\rangle$$

Use Hamiltonian $\hat{H}_1$ and $\hat{H}_2$ for the single-photon process and two-photon processes. For the vextor potential $\mathbf{A}$, $\hat{H}_1$ is the linear interaction term, proportional to $\mathbf{A} \cdot \mathbf{p}$. $\hat{H}_2$is the quadratic interaction term, proportional to $\mathbf{A} \cdot \mathbf{A}$

$$\hat{H}_1 = -\sum_j \frac{e_j}{m_j} \sum_{\nu,k} \sqrt{\frac{2\pi\hbar}{\omega_k V}} \left\{ \epsilon_k \cdot \mathbf{p}_j \; \hat{a}_k e^{i\mathbf{k}\cdot\mathbf{r}_j} + \epsilon_k \cdot \mathbf{p}_j \; \hat{a}_k^\dagger e^{-i\mathbf{k}\cdot\mathbf{r}_j} \right\} \tag{1}$$

$$\hat{H}_2 = \sum_j \frac{{e_j}^2}{m_j} \sum_{\nu,k} \sum_{\nu',k'} \frac{\pi\hbar}{\sqrt{\omega_k \omega_{k'} V^2}} \left\{ \epsilon_k \cdot \epsilon_{k'} (\hat{a}_k + \hat{a}^\dagger_{-k})(\hat{a}_{k'} + \hat{a}^\dagger_{-k'}) e^{i(\mathbf{k}+\mathbf{k}')\cdot \mathbf{r}_j} \right\} \quad (2)$$

ϵ_k is the polarization vector for the mode k.

In the interaction picture, the time evolution operator of the state vector is given by $|\alpha, t_0 : t\rangle = U_l(t, t_0) |\alpha, t_0 : t_0\rangle$. The Schrödinger equation in the interaction picture equation is equivalent with:

$$i\hbar \frac{\partial}{\partial t} U_I(t, t_0) = V_I(t) U_I(t, t_0)$$

According to the time-dependent perturbation theory with the probability amplitude of finding the system in state $|f\rangle$ at time t is expanded as:

$$\langle f| U_I(t, t_0) |i\rangle = c_f(t) = c_f^{(0)}(t) + c_f^{(1)}(t) + c_f^{(2)}(t) + \cdots \quad (3)$$

where:

$$c_f^{(0)}(t) = \delta_{fi}$$

Taken perturbation $V(t > 0) = V$ as a constant and $t_0 = 0$, we have:

$$\begin{aligned} c_f^{(1)}(t) &= \frac{1}{i\hbar} \int_{t_0}^{t} \langle f|V_I(t')|i\rangle dt' \\ &= \frac{1}{i\hbar} \int_{t_0}^{t} V_{fi}(t') e^{i\omega_{fi}t'} dt' \\ &= \frac{V_{fi}}{E_f - E_i} (1 - e^{i\omega_{fi}t}) \end{aligned} \quad (4)$$

$$\begin{aligned} c_f^{(2)}(t) &= \frac{1}{(i\hbar)^2} \sum_m \int_{t_0}^{t} dt' \int_{t_0}^{t'} dt'' e^{i\omega_{fm}t'} V_{fm}(t') e^{i\omega_{mi}t''} V_{mi}(t'') \\ &= -\frac{1}{i\hbar} \sum_m \frac{V_{fm} V_{mi}}{E_f - E_i} \int_0^t \left(e^{i\omega_{fi}t'} - e^{i\omega_{fm}t'} \right) dt' \end{aligned} \quad (5)$$

Define the state density of the final state $\rho(E)$ to be the number of states in $(E, E + dE)$. The probability of finding the system in the final state $|f\rangle$ at time t is given by:

$$\sum_f |c_f^{(1)}|^2 = \int dE_f \rho(E_f) |c_f^{(1)}|^2$$

$$= 4 \int \sin^2 \left(\frac{(E_f - E_i)t}{2\hbar} \right) \frac{|V_{fi}|^2}{(E_f - E_i)^2} \rho(E_f)\, dE_f$$

$$= \left(\frac{2\pi}{\hbar}\right) |V_{fi}|^2 \, \rho(E_f) t, t \to \infty, E_n \approx E_i$$

Consider the transition rate:

$$w_{i\to f} = \frac{d}{dt}(\sum_f |c_f^{(1)}|^2) = \left(\frac{2\pi}{\hbar}\right) |V_{fi}|^2 \, \delta(E_n - E_i)$$

This is called Fermi's Golden Rule. Use similar approach to analyze quadratic terms, we get the transition rate formula:

$$w_{i\to f} = \left(\frac{2\pi}{\hbar}\right) |V_{fi} + \sum_m \frac{V_{fm} V_{mi}}{E_f - E_i}|^2 \, \delta(E_n - E_i) \tag{6}$$

The first term is calculated to be:

$$V_{fi} = \frac{2\pi\hbar e^2}{mV} \sqrt{\frac{n_1(n_2+1)}{\omega_1 \omega_2}} \epsilon_1 \cdot \epsilon_2 \delta_{ca}$$

The second term involves two intermediated states $|b\rangle$, as shown in the figure.

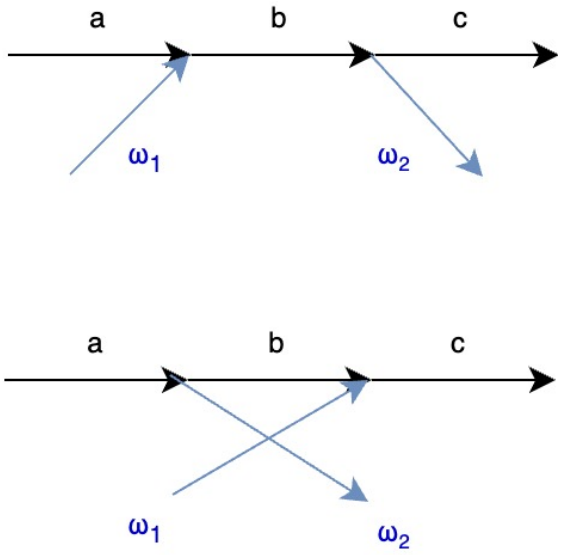


Figure 1: Two types of scattering due to two kinds of Hamiltonian.The first intermediate state is $|b\rangle \otimes |n_1 - 1, n_2\rangle$, the second is $|b\rangle \otimes |n_1, n_2 + 1\rangle$

The transition rate can be derived as:

$$w_{i\to f} = 2\pi\hbar \left(\frac{2\pi e^2}{mV}\right)^2 \frac{n_1(n_2+1)}{\omega_1\omega_2} \times \left| \epsilon_1\cdot\epsilon_2\,\delta_{ca} + \frac{1}{m}\sum_b \left(\frac{\langle c|\epsilon_2\cdot\hat{\mathbf{p}}|b\rangle\langle b|\epsilon_1\cdot\hat{\mathbf{p}}|a\rangle}{E_c - E_b + \hbar\omega_2} + \frac{\langle c|\epsilon_1\cdot\hat{\mathbf{p}}|b\rangle\langle b|\epsilon_2\cdot\hat{\mathbf{p}}|a\rangle}{E_c - E_b - \hbar\omega_1} \right) \right| \times \delta(E_a + \hbar\omega_1 - E_c - \hbar\omega_2) \tag{7}$$

The first term is the Rayleigh component since $E_c = E_a$. The contribution with $E_c \neq E_a$ is called the Raman component.

Equation (6) is called the Kramers-Heisenberg formula for steady-state light scattering.

We define $R_{21}^{ac}(\omega_1)$ as:

$$R_{21}^{ac}(\omega_1) = \frac{1}{m}\sum_b \left(\frac{\langle c|\epsilon_2\cdot\hat{\mathbf{p}}|b\rangle\langle b|\epsilon_1\cdot\hat{\mathbf{p}}|a\rangle}{E_b - E_c + \hbar\omega_1 + i\hbar\gamma} + \frac{\langle c|\epsilon_1\cdot\hat{\mathbf{p}}|b\rangle\langle b|\epsilon_2\cdot\hat{\mathbf{p}}|a\rangle}{E_b - E_a - \hbar\omega_1 + i\hbar\gamma} \right) \tag{8}$$

$$= \frac{e^2}{m}\sum_b \left(\frac{\langle c|\epsilon_2\cdot\hat{\mathbf{r}}|b\rangle\langle b|\epsilon_1\cdot\hat{\mathbf{r}}|a\rangle}{E_b - E_c + \hbar\omega_1 + i\hbar\gamma} + \frac{\langle c|\epsilon_1\cdot\hat{\mathbf{r}}|b\rangle\langle b|\epsilon_2\cdot\hat{\mathbf{r}}|a\rangle}{E_b - E_a - \hbar\omega_1 + i\hbar\gamma} \right) \tag{9}$$

which is called the Raman tensor.

$i\hbar\gamma$ introduces a small imaginary part because of the broadening in the intermediate state .

Then when $a \neq c$ we have:

$$w_{i\to f} = 2\pi\hbar \left(\frac{2\pi e^2}{mV}\right)^2 \frac{n_1(n_2+1)}{\omega_1\omega_2} \left|R_{21}^{ac}(\omega_1)\right|^2 \times \delta(E_a + \hbar\omega_1 - E_c - \hbar\omega_2) \tag{10}$$

Given two states with energy E_a and E_c, swapping the initial and final states gives two different ω_2s:

$$\omega_{21} = \omega_1 + \frac{E_c - E_a}{\hbar} \tag{11}$$

$$\omega_{22} = \omega_1 - \frac{E_c - E_a}{\hbar} \tag{12}$$

The two frequencies are called Anti-Stokes and Stokes lines, respectively.
Just by measuring the Stokes line in experiments, we can get theoretically energy differences between all combinations of the states of the system.

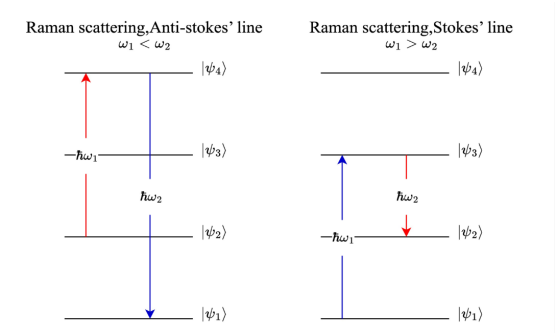


Figure 2: Raman Scattering

2.2 Intensity Ratio

Based on the Kramers-Heisenberg-Dirac (KHD) formula given in the previous section, if there is not much difference between ω_1 and ω_2,i.e. the vibrational frequency is much smaller than the frequency of the incident light, the transition rate will be approximatly the same. Then the intensity will only be dependent on the number of molecules on the state.
Let $\omega_\nu = \mid \omega_2 - \omega_1 \mid$ and substituting the Boltzmann distribution relation $\frac{N_1}{N_0} = e^{-\hbar\omega_v/k_BT}$, the intensity ratio between stokes and anti-stokes lines are as follow [16]:

$$\frac{I_{\text{Anti-Stokes}}}{I_{\text{Stokes}}} \approx e^{-\hbar\omega_v/k_BT}$$

Substituting the data,using C=O in CO2 as an example

- Vibration frequency: $\tilde{\nu} = 1388\,\text{cm}^{-1}$
- Excitation wavelength: $\lambda = 532\,\text{nm}$ ($\tilde{\omega} = 18\,796\,\text{cm}^{-1}$)
- Temperature: $T = 300\,\text{K}$

$$\begin{aligned} f_B &= \exp\left(-\frac{E_{\text{vib}}}{k_BT}\right) \\ &= \exp\left(-\frac{2.76\times10^{-20}\,\text{J}}{(1.38\times10^{-23}\,\text{J K}^{-1})(300\,\text{K})}\right) \\ &= e^{-6.67} \\ &= 0.00125 \end{aligned} \tag{13}$$

As the equation shows,the anti-stokes lines are far more weak than the stokes lines, however, the wavenumber shift from of the two kinds of lines are the same,which

indicates that they reflect the same molecular structural information.Thus, in experiments we can simply use the stokes lines to analyze the struct of the molecule. Given parameters:

- Differential scattering cross-section: $\dfrac{d\sigma}{d\Omega} = 10^{-30}\,\mathrm{cm}^2/\mathrm{sr}$
- Number density of particles: $N = 10^{22}\,\mathrm{cm}^{-3}$
- Medium length: $l = 1\,\mathrm{cm}$
- Detection solid angle: $\Omega = 0.1\,\mathrm{sr}$

Intensity ratio formula:

$$\frac{I_{\mathrm{sc}}}{I_0} = N \cdot l \cdot \frac{d\sigma}{d\Omega} \cdot \Omega \tag{14}$$

Numerical calculation:

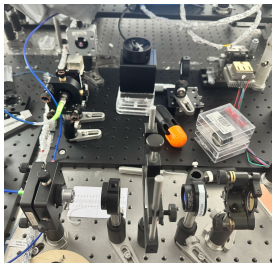
$$\begin{aligned} \frac{I_{\mathrm{sc}}}{I_0} &= \left(10^{22}\,\mathrm{cm}^{-3}\right) \cdot (1\,\mathrm{cm}) \cdot \left(10^{-30}\,\mathrm{cm}^2/\mathrm{sr}\right) \cdot (0.1\,\mathrm{sr}) \\ &= 10^{22} \cdot 1 \cdot 10^{-30} \cdot 0.1 \\ &= 10^{-9} \end{aligned} \tag{15}$$

This indicates that the intensity of the scattered light is significantly weaker than the incident light. If we are to obtain recognizable spectral lines, we need to use a laser source with power of at least 100 mW.

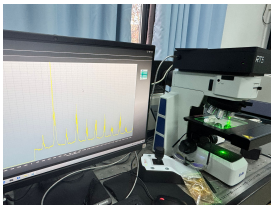
3 Data Acquisition and Analysis

We conducted several pre-experiments with DIY Raman spectrometer. However,we discovered that the wave shifts are too weak to be distinguished from background noise.This is because the scattered light that collected by the spectrometer is not enough. To address this problem,we built an improved spectrometer, as shown in 3a. A notch filter is also added to better distinguish the signal from the incident light.

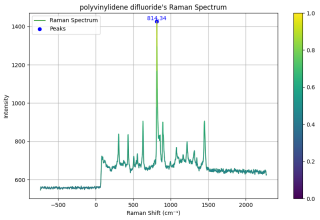
To obtain high-quality spectral data, we utilized a commercial spectrometer to measure the Raman spectra of various organic polymers. The spectrometer, equipped with a microscope, efficiently collects scattered light from all directions while utilizing filters to eliminate stray light, enabling us to acquire high-resolution spectral data.



(a) The improved spectrometer which work on 1064 nm laser.



(b) Zolix spectrometer which work on 532 nm laser.



(c) Example of experimental data (spectrum of polyvinylidene difluoride).

We obtained 50+ spectrum of polymers and non-polymers and compared them in different aspects. We have found three iconic patterns,w which will be explained in the following subsections.

3.1 Functional Groups

Organic compounds with similar functional groups exhibit peaks in the same region. That's because certain bonds have identical energy levels, which is revealed in the pattern of the spectra. We designed a simple algorithm to find the characteristic peaks with the spectral of a group of chemicals with the same chemical group.

Given a set of N normalized continuous functions $S_i(x)$ [17], where $S_i(x) \in [0, 1]$ for all i and x, we define their product spectrum as:$P(x) = \prod_{i=1}^{N} S_i(x)$.

Theorem: Let N continuous functions $S_i(x) \in [0, 1]$ have local maxima at x_0 with $S_i(x_0) \geq \theta$. Then, their product $P(x)$ has a local maximum at x_0 and has value $< \theta^N$ elsewhere, with selectivity increasing as N grows.

Based on this theorem, the pipeline is as follows:

1. Normalization
2. Interpolation and Smoothing
3. Product Calculation
4. Peak-Finding

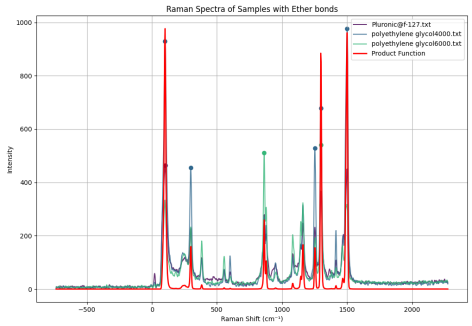


Figure 4: Spectrum of polymers Pluronic@f-127, polyethylene glycol4000 and polyethylene glycol6000 (all with ether bonds) and the common characteristic extraction (shown in red).

3.2 Degree of Polymerization

We have found that the degree of polymerization (DP) can affect the relative intensity of Raman characteristic peaks, but not the position of the peaks. This is because of the long-range interaction of the bonds.

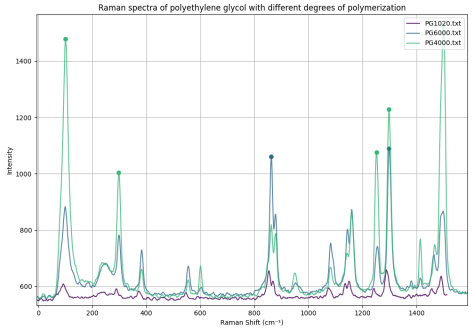


Figure 5: Spectrum of polymers polyethylene glycol 1020, 4000 and 6000. The numbers indicate their degree of polymerization.

3.3 Solution

The Raman spectrum of a solution is the sum of the contributions from the solute and the solvent, as shown in Fig.8.

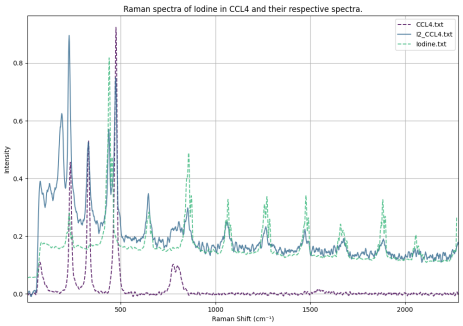


Figure 6: Spectrum of Iodine in CCl4. The component spectra are shown in dashed lines.

4 Integrated Identification Method

Based on the above analysis, We designed a pipeline to identify organic compounds, taking the QM9S dataset [18] as an example. The first step is to get a group of candidates with similar spectral pattern, that is to say, have peaks in the same region. We conducted the experiment on An integrated method that employs vector embedding based on the position of peaks and the k-nearest neighbors (k-NN) algorithm can do the job, as shown in Fig.9.

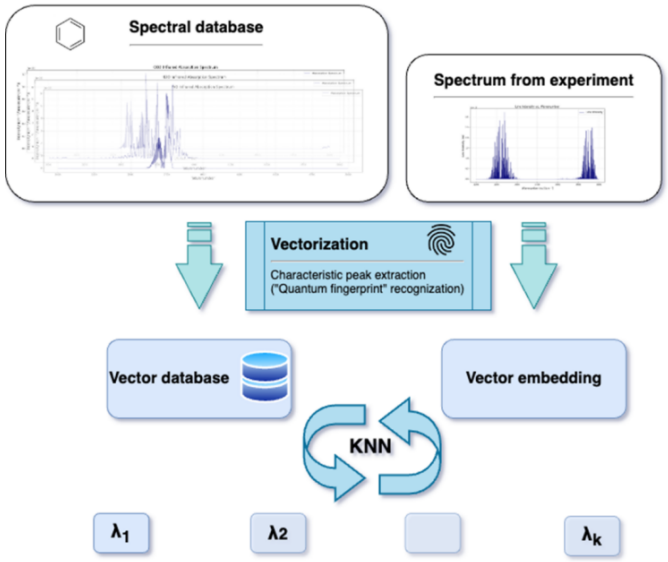


Figure 7: Candidate selection from a spectral database.

The proposed methodology can be compactly represented by the following unified formulation:

In this step, we leverage the Principal Components Analysis (PCA) [19] for feature extraction and dimension reduction. The dimensionality of the spectral data is reduced from 3501 to 400 dimensions, which speeds up the search process by 875%.

$$\mathbf{S} = \underset{j\in\{1\cdots N\}}{\text{argtopS}} \left\{ \sum_{k=1}^{d} \| (\frac{x_{\text{ref}}}{\max(x_{\text{ref}})})\mathbf{W} - (\frac{x_j}{\max(x_j)})\mathbf{W} \|_1 \right\} \tag{16}$$

where:

- x_{ref} and x_j are the reference and query spectra.
- $\mathbf{W}$ contains the PCA eigenvectors (top d components). Here $d = 400$.
- argtopS selects the S spectra with smallest L1 distances.
- N is the total number of spectra in the dataset. For

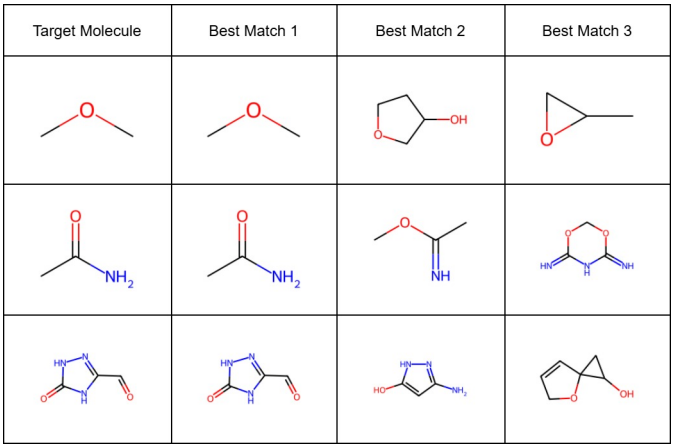


Figure 8: Candidate selection in the PCA reduced QM9S dataset.

Note that there is an interesting phenomenon that molecular with similar structures tend to cluster together in the PCA space. After the candidates are selected, the problem converts to finding the most possible structure from the information of raman spectrum. For human experts, there are a pile of empirical rules to decide if there are certain structures in the molecule. [20] These rules specifies that if there is a certain chemical bond or functional groups in a molecule, then there will be a peak in certain range. These rules are given to the AI agents to help them reason about the structure of the molecule.

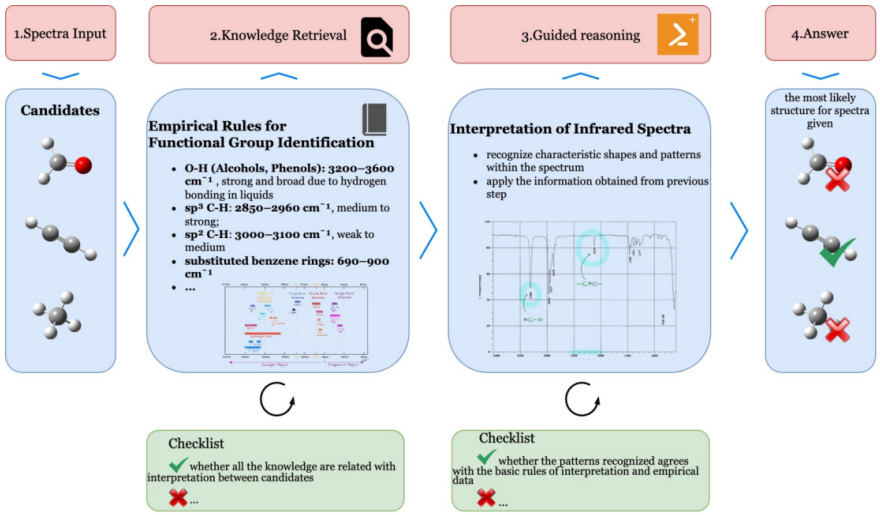


Figure 9: Matching between molecular spectra and their corresponding structures utilizing the reasoning ability of LLMs.

Here's an example:

- A strong peak at 1700 to 1750 cm^{-1} indicates the presence of a carbonyl (C=O) group.
- A broad peak at 3200 to 3600 cm^{-1} corresponds to the stretching vibration of hydroxyl (OH) groups.
- Aromatic rings exhibit a distinctive triplet of peaks at 1580, 1600, and 1500 cm^{-1}.

The workflow shown in Figure 9 is as follows:

1. **Knowledge Embedding:** More than 200 spectral-structural association rules from Ref. [20] were encoded into prompt templates.
2. **Multimodal Input:** Candidate molecular structures (expressed in SMILES format) and the measured Raman peak positions (as shown in Fig. **??**) were provided as input.
3. **Reasoning Process:** The LLM employed chain-of-thought reasoning

After each step, the LLM is guided to check its output to avoid fallacious implementations of the knowledge base.

We applied the above method to three State-of-the-Art LLMs: ChatGPT-4o, Deepseek-r1 and ChatGPT o3. The results are shown in Table below.

Method	**GPT-4o**	**deepseek-r1:671B**	**GPT-o3**
CoT(system)	0.0	16.7	0.0
KNN	16.7	**50.0**	<u>50.0</u>
Fingerprint Reasoner (Ours)	**50.0**	<u>33.3</u>	**50.0**

Table 1: Accuracy results (%) on spectroscopic reasoning problems. **Boldface** numbers highlight the best results; <u>underlined</u> numbers represent the second-best.

We witness a general improvement in the identification process. However, the improvement is greater for LLMs without reasoning framework.

5 Conclusion

This study establishes a comprehensive theoretical, experimental, and algorithmic framework for identifying organic compounds via Raman spectroscopy. Quantum mechanical analysis indicates that molecular vibrational energy level differences

determine the Stokes and anti-Stokes shift positions, while the bond polarizability tensor R_{21}^{ac} governs peak intensities, thereby providing a physical basis for spectral fingerprinting. Experimental findings include:

1. **Functional group characteristic peaks** exhibit strong specificity (e.g., COC stretching vibrations for ether bonds at 800 to 1200 cm $^{-1}$).
2. **Degree of polymerization** affects peak intensity without shifting peak positions (due to long-range interactions).
3. **Solution spectra** follow the principle of linear superposition.

The proposed integrated algorithmic framework overcomes limitations of conventional methods:

- A **PCA-kNN filter** reduces spectral data into a 90-dimensional feature space and performs dot-product similarity matching against the *qm9s* database:
- A **LLM reasoning engine** executes chain-of-thought reasoning (spectral features $\rightarrow$ functional groups $\rightarrow$ spatial matching), achieving accurate decisions.

This method offers both **precise identification within existing databases** and **structural inference for unknown compounds**. Its current limitations lie in the completeness of the reference spectral database. Future work will integrate multimodal data such as infrared and mass spectrometry, and develop dedicated spectral-to-structure transformation large models to advance intelligent analytical chemistry.

6 Acknowledgements

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7 appendix

The following is an example of the spectral-to-molecule matching process completed by ChatGPT-4o, showing the complete prompt and output:

Problem Description

An organic compound has Raman peaks at:
483, 495, 515, 600, 627, 654, 674, 779, 813, 869, 934, 1014, 1040, 1115, 1135, 1224, 1350, 1381, 1524, 1568, 1600 cm^{-1}
Possible choices: **polyethylene glycol**, **citric acid**, **glucose**

Known Raman Peaks for Candidates

Compound	Peak Positions (cm^{-1})	Vibrational Modes
PEG	848, 1060, 1146, ...	C-O-C stretching, CH_2 vibrations
Citric acid	486, 516, 605,...	O-H bending, C=O stretching, C-C stretching
Glucose	478, 519, 610, ...	C-O stretching, C=C stretching, CH vibrations

Peak Matching Analysis

Peak (cm^{-1})	PEG	Citric Acid	Glucose
483	No	Yes	Yes
495	No	Yes	Yes
515	No	Yes	Yes
600	No	Yes	Yes
627	No	Yes	No
654	No	Yes	No
674	No	Yes	No
779	Yes	Yes	Yes
813	No	No	No
869	No	Yes	Yes
934	No	Yes	Yes
1014	No	Yes	Yes
1040	No	No	No
1115	No	Yes	Yes
1135	No	Yes	No
1224	No	Yes	No
1350	No	No	No
1381	No	Yes	No
1524	No	Yes	No
1568	No	Yes	No
1600	No	Yes	No

Conclusion: Citric Acid Identification

The observed Raman peaks align perfectly with the characteristic signature of **citric acid**, notably at: 483, 515, 600, 674, 779, 869, 934, 1014, 1115, 1224, 1381, 1524, 1568, and 1600 cm$ˆ-1$. Key evidence includes:

- 14 out of 20 peaks match citric acid (compared to 7 for glucose).
- Distinctive carboxylic acid peaks at 15241600 cm^{-1}.
- Unique matches at 627, 654, 674, 1135, and 1224 cm^{-1}.

Final Identification: Citric Acid

This demonstrates that the proposed method successfully performed fuzzy matching between spectral samples and molecular structures, achieving the intended identification objective.

Prompt Templates for Baselines	
Sys	You are an expert in chemistry and quantum physics. Please provide a clear and step-by-step solution for a spectroscopic problem. An organic compound has raman peaks at 483 cm^-1 495 cm^-1 515 cm^-1 600 cm^-1 627 cm^-1 654 cm^-1 674 cm^-1 779 cm^-1 813 cm^-1 869 cm^-1 934 cm^-1 1014 cm^-1 1040 cm^-1 1115 cm^-1 1135 cm^-1 1224 cm^-1 1350 cm^-1 1381 cm^-1 1524 cm^-1 1568 cm^-1 1600 cm^-1. What is it? { Example Answer} Step one, list the given Raman peaks and recognize that several fall within regions typically associated with skeletal deformations, carbon-hydrogen bending, methyl group vibrations, and aromatic carbon-carbon double bond stretches. Step two, notice the peaks around 1524, 1568, and 1600, which are characteristic of aromatic ring stretching, strongly indicating an aromatic compound. Step three, observe the peaks at 1350 and 1381, which suggest the presence of a methyl group attached to the aromatic ring. Step four, compare the overall pattern to known reference spectra, finding a close match to toluene, which is methylbenzene, based on the combination of aromatic and methyl-related vibrations. Therefore, the compound is toluene.
KNN	An organic compound has raman peaks at 483 cm^-1 495 cm^-1 515 cm^-1 600 cm^-1 627 cm^-1 654 cm^-1 674 cm^-1 779 cm^-1 813 cm^-1 869 cm^-1 934 cm^-1 1014 cm^-1 1040 cm^-1 1115 cm^-1 1135 cm^-1 1224 cm^-1 1350 cm^-1 1381 cm^-1 1524 cm^-1 1568 cm^-1 1600 cm^-1 what is it? Possible choices are polyethylene glycol, citrus acid and glucose. { Example Answer} Step one, list the given Raman peaks and recognize that several fall within regions typically associated with skeletal deformations, carbon-hydrogen bending, methyl group vibrations, and aromatic carbon-carbon double bond stretches. Step two, notice the peaks around 1524, 1568, and 1600, which are characteristic of aromatic ring stretching, strongly indicating an aromatic compound. Step three, observe the peaks at 1350 and 1381, which suggest the presence of a methyl group attached to the aromatic ring. Step four, compare the overall pattern to known reference spectra, finding a close match to toluene, which is methylbenzene, based on the combination of aromatic and methyl-related vibrations. Therefore, the compound is toluene.

Table 2: Prompt templates for baselines.

Prompt Templates for Fingerprint Reasoner	
Knowledge Retrieval	Let's think step by step: 1.You are an expert in chemistry. Given the three candidate molecules, find all the bonds in the molecule. Find the feature regions in the reference table given below. { RAMAN Band Correlation Table}
Checklist 1	2.You are a careful student. Now you have retrieved known vibrational modes of these three candidates. Check whether all the knowledge you have retrieved are related with interpretation between candidates. Also check if you have found all the features, and every feature must come from the table. { Retrieved Features}
Correlation Analysis	3.Match the Raman peaks to the known vibrational modes of these three candidates.You can make a table to show the match. The table should include the Raman peak, the known vibrational mode, and the corresponding candidate molecule. { Correlation Table} 4. Find the candidate molecule with the most matches.
Checklist 2	5.Double check your answer. Check whether the patterns recognized agrees with the casic rules of interpretation and empirical data. If you are confident, you can submit your answer.

Table 3: Prompt templates for Fingerprint Reasoner (Ours).